

Security and Compliance

Use **VCF Operations** as a VI administrator to evaluate the security stance of your organization across the VCF fleet. Compliance management continuously monitors your infrastructure against industry standards and custom policies. It detects violations, highlights risk areas, and provides actionable remediation recommendations

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Getting to Know VCF

To apply this documentation, you must be acquainted with the [VCF Product Overview](#), [VCF Design](#), and [VCF Release Notes](#).

SDKs, APIs, and CLI for VCF Administration

You can build, operate, and manage your VCF private cloud by using the VCF SDK and APIs, and VCF PowerCLI. See [Administration SDKs, APIs, and CLI](#).

Monitoring Security Operations

Use VMware Cloud Foundation (VCF) Security Operations to view a consolidated, high-level, strategic view of the VCF fleet's security posture across the VMware Cloud Foundation platform. VCF Security Operations helps security stakeholders gain enhanced visibility and proactively plan for effective risk management.

What is Security Operations in VMware Cloud Foundation ?

VCF Security Operations aims to tackle the challenges associated with security management and its impact on operations by empowering administrators to proactively manage and improve the security posture of their VMware Cloud Foundation platform.

With this in mind, the main goals of VCF Security Operations are:

- Provide timely and accurate information in a centralized location to help you understand and mitigate potential/known security risks.
- Provide comprehensive visibility into security across multiple areas including user security and VCF Operations component security including VMs, ESX hosts, and vSAN.

Where You Find VCF Security Operations

From the VCF Operations left menu, click **Security > Security Operations** to access the **Security Operations** page. The **Security Operations** page provides a holistic view of the organization's user and infrastructure security stance across the VMware Cloud Foundation platform.

User Security

Discovering authentication trends like total logins, failed logins, and changes in user privileges within vCenter instances is crucial for maintaining the security and operational integrity of the VMware Cloud Foundation platform.

By continuously monitoring user security aspects, administrators can detect security risks early, respond to potential threats, and ensure that the right users have the right level of access at all times. VCF Security Operations provides insights into the security and operational health of the VMware Cloud Foundation platform.

Prerequisites

- To view authentication and user permission events across vCenters, configure integration with VCF Operations for logs. Ensure that you are using the following versions:
 - VCF Operations 9.0
 - VCF Operations for logs 9.0

Logs Integration

- For vSphere Foundation deployments, integrate VCF Operations for logs into VCF Operations. For more information, see [VCF Operations for logs for VVF clients](#).
- For VCF deployments, integrate VCF Operations for logs with VCF Operations. For more information, see [Deploying the VCF Operations for Logs Appliance](#).

Enable Log Collection

- Enable log collection from the vCenter adapter instance in the VCF Operations console. For more information, see [Configure a vCenter Account in VCF Operations](#).
- Enable log collection from domains of the VMware Cloud Foundation adapter instance. For more information, see [Configuring VCF Cloud Account in VCF Operations](#).
- Ensure that the requisite permission has been assigned to the user using log integration. From the VCF Operations console, navigate to **Administrator > Control Panel > Access Control > Roles** to assign the following requisite permission:
 - Security > Security Operations

Authentication

You can use the **Authentication** widget to discover user authentication trends on vCenter and discover anomalies. Using this widget, you can dive deeper into authentication records to understand who logged in into the system, when, and how.

Total Logins: Tracking how many users are logging into vCenter across different times or periods. This helps identify usage patterns, peak activity times, and potential unauthorized access attempts.

Failed Logins: Monitoring failed login attempts can reveal potential security threats, such as brute-force attacks or users with incorrect credentials. Analyzing the frequency and source of failed logins can guide in identifying issues and improving security measures.

To view authentication details, from the VCF Operations left menu, click **Security > Security Operations**. From the **Authentication** widget, click **View Details**. The events displayed are vCenter logs collected by VCF Operations for logs.

At the top of the **Authentication** page, you see the number of vCenter instances monitored. A trend line displays user authentication details over time. A list of authentication events is displayed in a data grid on the page. You can filter the authentication events by using the filter options.

Filter and Data Grid

You can filter user authentication events by authentication status, updated by, vCenter, object type, event, and description. Based on your filter, you can view user authentication details in the data grid such as user name, successful or failed authentications, the name of the user, events such as user login, guest operation authentication, and so on. For a detailed view of each authentication event, click on the double arrow in the data grid against each row.

User Permissions

You can use the **User Permissions** widget to understand which users have their roles or permissions changed and discover if those changes were legitimate. You can dive deeper to understand when such changes were made and who

made the change. These details are useful to trace any unauthorized changes. By tracking when new permissions are granted, modified, or revoked, you can get an understanding of who has access to what resources in vCenter.

To view User Permission details, from the VCF Operations left menu, click **Security > Security Operations**. From the **User Permissions** widget, click **View Details**. The user permissions events displayed are vCenter logs collected by VCF Operations for logs.

Filter and Data Grid

You can filter user permission events by event, status, description, updated by, object name, object type, and description. Based on your filter, you can view user permission details in the data grid. For example:

- Events, which could be permission updated or permission canceled.
- Status, such as created, updated, or deleted user permissions.
- The name of the user who created the user event, description, time stamp, and so on.

For a detailed view of each user permission event, click on the double arrow in the data grid against each row.

Time Range for Authentication and User Permissions Events

By default, VCF Operations displays authentication and user permission details for the last 24 hours. You can change this time range by clicking the drop-down menu. The presets available are:

- Last 1 hour
- Last 6 hours
- Last 12 hours
- Last 24 hours
- Last 48 hours

If you want to select a custom range, use the Start date, End date, Start time and End time options. The custom range is limited to a 48 hour period. Click **Apply** to filter the authentication events.

Infrastructure Security

For an immediate view of the security posture of your VMware Cloud Foundation platform, you can discover security advisories, host encryption modes, host modes, VM and vSAN cluster encryption, and certificates across VCF Instances.

Advisories

As part of the infrastructure security view, use the **Advisories** widget to view and access vSphere security advisories. These advisories are categorized by severity such as critical, immediate, or warning so that you can prioritize which ones to address

To view security advisories, from the VCF Operations left menu, click **Security > Security Operations**. From the **Advisories** widget, click **View Details** for detailed information on Diagnostic Findings. For more information, see the [What are Diagnostic Findings](#).

Note:

The **View Details** link is not available if there is no data.

Host Encryption Mode

Use the **Host Encryption Mode** widget to quickly identify ESX hosts that are not prepared and ready to carry out encryption tasks such as encrypting a VM disk. ESX hosts must be prepared to participate in security capabilities of the infrastructure.

To view host encryption modes, from the VCF Operations left menu, click **Security > Security Operations**. From the **Host Encryption Mode** widget, click **View Details**.

At the top of the **Host Encryption Mode** page, you see the total number of ESX hosts deployed across the data center. A pie chart displays the encryption modes of the ESX hosts. For information about host encryption modes, see [Using Encryption in your vSphere Environment](#). A list of ESX hosts and their encryption modes is displayed in a data grid on the page. You can filter the host encryption modes using the filter options.

Filter

You can filter host encryption modes by the host name, status, and vCenter. Based on your filter, you can view ESX hosts that are incapable, pending incapable, prepared, or safe.

Host Mode

Use the **Host Mode** widget to monitor and track the status of the ESX hosts. Knowing an ESX hosts status is crucial to ensure that adequate levels of security are applied at all times. The ESX host statuses other than Active, undermines the availability of your infrastructure and could potentially disrupt your production workloads. From this widget you can discover ESX hosts deployed across VCF Instances that require compensating security controls as they are in a non-active mode. You can also discover ESX host maintenance status across the data center.

To view host modes, from the VCF Operations left menu, click **Security > Security Operations**. From the **Host Mode** widget, click **View Details**.

At the top of the **Host Mode** page, you see the total number of ESX hosts deployed across the data center. A pie chart displays the modes of the ESX hosts. A list of ESX hosts and their modes is displayed in a data grid on the page. You can filter the host modes using the filter options.

Filter

You can filter host modes by the host mode, status, and host name. Based on the filter you set, you can view ESX hosts in different modes such as active, in quarantine, and in maintenance.

vSAN Cluster Encryption

Use the **vSAN Cluster Encryption** widget for an overview about the encryption status on different vSAN clusters.

To view vSAN cluster encryption, from the VCF Operations left menu, click **Security > Security Operations**. From the **vSAN Cluster Encryption** widget, click **View Details**. For more details see the [Monitoring Storage Operations](#).

Certificates

Use the **Certificates** widget for for an overview about certificates that require attention based on their expiration date.

To view certificates, from the VCF Operations left menu, click **Security > Security Operations**. From the **Certificates** widget, click **View Details**. For more details see [Managing Certificates in VMware Cloud Foundation](#).

VM Encryption

Use the **VM Encryption** widget to find out if your VM workloads are encrypted. You can find details about the storage policy, at per disk level, attached to the VM. If VMs are not encrypted, you should consider encrypting them to avoid data exposure.

To view VM encryption details, from the VCF Operations left menu, click **Security > Security Operations**. From the **VM Encryption** widget, click **View Details**.

At the top of the **VM Encryption** page, you see the total number of VMs deployed across the data center. A pie chart displays the encryption status of the VMs. Encryption status could be encrypted, partially encrypted, or unencrypted.

Filter and Data Grid

You can filter VMs by encryption status, vCenter name, host name, and VM name. You can view the disk details and storage policy attached to each VM by clicking the **View Details** link under the **VM Disk Details** column. This allows you to view the number of disks in the VM and determine whether they are encrypted. For example, if the VM encryption status is partially encrypted and the VM has five disks, you see that some of the disks are not encrypted and some are encrypted.

Viewing and Configuring Compliance

VCF Operations helps you visualize the compliance of the objects in your inventory by displaying compliance benchmarks which are measured against a set of standard rules, regulatory best practices, or custom alert definitions. You can discover non-compliant objects and the associated compliance violations and take remedial measures.

Measuring Compliance of Objects

VCF Operations continuously monitors your infrastructure to ensure that it remains in compliance. Compliance is an ongoing process. VCF Operations evaluates the collected data against the defined policies. It assigns a compliance score to each object or group based on how well they adhere to the policies. The compliance score is typically represented as a percentage.

How Compliance Benchmarks Work

Compliance is used to monitor the vCenter instances, vSAN, NSX and Cloud Environments. Objects which can be monitored include the hosts, virtual machines, distributed port groups, and distributed switches in your environment. Compliance benchmarks helps to ensure that the settings on your objects meet the defined standards. Score cards help you proactively detect compliance problems in objects managed by VCF Operations. The compliance benchmarks are measured against a set of standard rules, regulatory best practices, or custom alert definitions.

All the compliance standards in VCF Operations, including any standards that you define, are based on alert definitions. Only alert definitions of the Compliance subtype are counted. Custom score cards can monitor user-defined alerts. The alerts and symptom definitions are based on the properties and metrics of the underlying object.

When VCF Operations detects non-compliance with a policy, it can generate alerts or notifications. Depending on the severity of the non-compliance, you can configure automated remediation actions to bring the object back into compliance.

You can manage all compliance related tasks from the **Security > Compliance** page. The data sources are displayed on the top of the page. To see a compliance score card, you must first configure the data source that VCF Operations can monitor, and then activate the benchmarks for those types of data sources. When you activate a benchmark for a data source, you select an applicable policy. VCF Operations then activates the appropriate alert definitions in the policy to measure compliance.

Data Sources for Calculating Compliance

VCF Operations can measure compliance from two data sources to ensure that your virtualized environment adheres to predefined policies and standards. These data sources provide the information necessary to evaluate the compliance of your infrastructure. The data sources that VCF Operations can use for compliance measurement are as follows:

- vCenter
- VMware Cloud Foundation

Note:

For VCF Operations to measure compliance against these data sources, you must first configure them. See the relevant topic in [Integrating Data Sources](#).

Compliance Benchmarks

VMware SDDC Benchmarks

Displays score cards based on alerts which are measured against the latest hardening guides:

- vSphere Security Configuration Guide
- vSAN Security Configuration Guide
- NSX Security Configuration Guide

For more details, see [VMware SDDC Benchmark Details](#).

Note:

In future versions of VCF Operations, these benchmarks will be deprecated. A new, consolidated, out of the box benchmark called, "VCF 9 Security Baseline" will be introduced.

Displays score cards based on alerts for VMware Cloud Foundation which are measured based on the following:

- VCF 4.2
- VCF 4.3
- VCF 4.4
- VCF 4.5
- CIS Compliance Pack for VCF
- VCF Operations CSA Compliance Pack for VCF.

Note:

In future versions of VCF Operations, these benchmarks will be deprecated. A new, consolidated, out of the box benchmark called, "VCF 9 Security Baseline" will be introduced.

Note:

You must install and activate the Aria Operations CSA Compliance Pack for VCF after downloading the .PAK file from [Marketplace](#).

The alerts are based on objects in your VMware Cloud Foundation platform.

For more details, see the topic, [VCF Benchmarks based on VMware Cloud Foundation Security Baseline](#).

Custom Benchmarks

Displays benchmarks that you define. Use compliance alerts from integrations, and regulatory management packs, or define your own alerts to monitor. You can define up to five custom score cards. You can import custom benchmarks from other instances of VCF Operations.

Regulatory Benchmarks

Displays benchmarks for industry standard regulatory compliance requirements. You can enable or install compliance packs for the following regulatory standards:

- Defense Information Systems Agency (DISA) Security Standards. This is available from the repository as a native pack.
- The Federal Information Security Management Act (FISMA) Security Standards. This is available from the repository as a native pack.
- Health Insurance Portability and Accountability Act (HIPAA). This is available from the repository as a native pack.
- International Organization for Standardization (ISO) Security Standards. This is available from the repository as a native pack.

- Payment Card Industry Data Security Standard (PCI DSS) Compliance Standards
- CIS Security Standards. Install this pack from the VMware Marketplace.
- NIST SP 800-171, Protecting Controlled Unclassified Information in Nonfederal Systems and Organizations. Install this pack from the VMware Marketplace.
- NIST SP 800-53 R5. Install this pack from the VMware Marketplace.

For more details, see [Regulatory Benchmark Details](#).

Note:

In future versions of VCF Operations, these benchmarks will be deprecated. A new, consolidated, out of the box benchmark called, "VCF 9 Security Baseline" will be introduced.

For instructions on activating these compliance packs, see [Activate a Regulatory Benchmark](#).

VCF Benchmarks based on VMware Cloud Foundation Security Baseline

Security and Compliance for VCF provides guidance for auditors who would want to evaluate their VCF environment. You can leverage the benchmarks and regulatory frameworks available under VCF based on needed benchmarks or regulatory frameworks of your choice.

VCF Security Baseline

Security Baseline for VCF is a document that builds on top of VCF and leverages security fundamentals. The spreadsheet addresses the most frequently requested compliance standards, regulations, and frameworks.

The security baseline is designed and validated to tailor security configurations without impacting the ability of VCF to meet its design objectives. The security baseline can assist organizations to secure information systems in a compliance context.

The VCF security baseline can be used to evaluate both default and non-default configurations. For more information, see [Regulatory Compliance Information](#) on GitHub.

For information about the compliance conditions required for assessing compliance of VCF environments, download the file: [VCF Audit Guide](#).

Versions of the supported VCF security baseline are:

- VCF 4.2
- VCF 4.3
- VCF 4.4
- VCF 4.5

VCF Operations supports the following products for VCF Benchmarks:

- ESX
- SDDC Manager,
- vCenter
- vSAN
- NSX

Within these products, VCF Operations can measure the compliance of the following resources:

- VMware Distributed Virtual Switch

- VMware Adapter Instance
- Virtual Machine
- Host System
- Logical Router
- NSX
- NSX Manager Service
- Logical Switch
- Management Cluster
- VCF Adapter Instance
- vSAN
- Data Center
- ESX Cluster

Note: Assessment of objects starting from version VCF 4.5 and above is done based on VCF 4.5 security baseline.

Note:

For VCF Operations to measure compliance against these data sources, you must first configure them. See the relevant topic in [Integrating Data Sources](#).

VMware Aria Operations CSA Compliance Pack for VCF

The Cloud Security Alliance (CSA) framework holds significant importance within the sphere of cloud computing, offering comprehensive guidance and best practices for ensuring the security of cloud-based systems and services.

The Aria Operations CSA Compliance Pack for VCF is available as a Management Pack. Before you can enable the Aria Operations CSA Compliance Pack for VCF, you must install and activate the integration. Download the Management Pack from the [VMware Marketplace](#). For information on how to install a management pack, see the topic, [Adding and Managing Management Packs from VMware Marketplace](#).

For more information about the control set, download the file: [CSA Controls](#).

CIS Compliance Pack for VCF

The CIS (Center for Internet Security) Security Standards are a set of best practices and guidelines for securing computer systems and networks. The CIS organization is a non-profit entity that collaborates with experts from various industries to develop and promote consensus-based security configurations and benchmarks.

The supported version is CIS VMware ESXi 8.0 Benchmark v1.1.0 Dated 03-25-2024. For more information about the control set, download the file: [CIS for VCF](#).

VMware SDDC Benchmark Details

VMware Software-Defined Data Center (SDDC) security benchmarks are guidelines or standards that provide recommendations and best practices for securing VMware's SDDC infrastructure. These benchmarks help organizations assess and enhance the security of their virtualized data center environments.

Note:

In future versions of VCF Operations, these benchmarks will be deprecated. A new, consolidated, out of the box benchmark called, "VCF 9 Security Baseline" will be introduced.

vSphere Security Configuration Guide

For details of the conditions that allow automated compliance assessment and the list of controls that can be used to perform manual checks based on the VMware vSphere 7 Security Configuration Guide Update 3 and VMware vSphere 8 Security Configuration Guide. For more information, download the files:

- [vSphere 7 Security Configuration Guide Update 3](#) .
- [vSphere 8 Security Configuration Guide](#)

The vSphere Security Configuration Guide is a comprehensive document provided by VMware that outlines best practices and recommendations for securing VMware vSphere, which is a virtualization platform used for creating and managing virtualized data centers.

The guide covers various aspects of vSphere security and provides configuration guidelines to help administrators protect their vSphere infrastructure from potential threats and vulnerabilities. It addresses security considerations for different components of the vSphere environment, including ESX hosts, vCenter, virtual machines, networking, and storage.

The vSphere Security Configuration Guide typically covers the following areas:

- **ESX Host Hardening:** It provides recommendations for securing the ESX hosts, including configuring authentication and authorization settings, disabling unnecessary services, enabling security features like Secure Boot and Trusted Platform Module (TPM), and implementing network security measures.
- **vCenter Hardening:** The guide offers best practices for securing the vCenter, which is a centralized management platform for vSphere. It covers securing the vCenter installation, securing access and authentication, configuring roles and permissions, and enabling auditing and logging.
- **Virtual Machine Security:** It provides recommendations for securing virtual machines (VMs), such as implementing secure VM configurations, using virtual machine encryption, enabling guest operating system security features, and protecting VMs from unauthorized access.
- **Networking Security:** The guide offers guidelines for securing the vSphere networking infrastructure, including configuring virtual switches, VLANs, and firewalls, implementing network segmentation, and securing network communication between vSphere components.
- **Storage Security:** It covers best practices for securing the vSphere storage environment, including securing access to storage devices, implementing storage encryption, and protecting virtual machine data.
- **Monitoring and Auditing:** The guide emphasizes the importance of monitoring and auditing vSphere components, and provides recommendations for enabling and configuring logging, using security information and event management (SIEM) tools, and monitoring for suspicious activities.

The vSphere Security Configuration Guide is regularly updated by VMware to incorporate the latest security recommendations and considerations. It serves as a valuable resource for administrators seeking to secure their vSphere infrastructure and protect their virtualized environments from potential security risks.

vSAN Security Configuration Guide

The vSAN Security Configuration Guide is a document provided by VMware that offers guidance and best practices for securing vSAN, which is a software-defined storage solution integrated into the vSphere environment.

The guide focuses on helping administrators secure the vSAN infrastructure and protect the data stored within it. It provides recommendations for various security considerations related to vSAN, including authentication, encryption, network security, and access controls.

Here are some key areas covered in the vSAN Security Configuration Guide:

- **Authentication and Authorization:** The guide provides recommendations for configuring strong authentication mechanisms for vSAN components, such as vCenter and ESX hosts. It suggests using secure protocols, enforcing secure password policies, and implementing multi-factor authentication where possible. It also covers role-based access control (RBAC) and permissions management.

- **Encryption:** vSAN supports encryption at rest, which ensures that data stored on vSAN disks is protected. The guide offers guidance on configuring vSAN encryption, including selecting the appropriate encryption algorithm, managing key management servers (KMS), and enabling encryption for data at rest.
- **Network Security:** It provides recommendations for securing network communication within the vSAN environment. This includes using secure protocols, enabling network encryption (VMkernel Encryption), and configuring firewall rules to control network traffic between vSAN components.
- **Auditing and Logging:** The guide emphasizes the importance of monitoring and auditing vSAN operations. It offers recommendations for enabling and configuring logging, implementing log management solutions, and monitoring vSAN events for potential security incidents.
- **Secure Configuration Settings:** It provides a list of vSAN-specific configuration settings that can enhance security. This includes recommendations for enabling features like Data-at-Rest Encryption, Secure Erasure, and ensuring proper network configurations.
- **Compliance and Hardening:** The guide addresses compliance considerations and provides information on how vSAN aligns with various security frameworks, such as the Center for Internet Security (CIS) benchmarks. It also offers recommendations for hardening vSAN components, including ESX hosts and vCenter.

The vSAN Security Configuration Guide is regularly updated by VMware to incorporate the latest security best practices and recommendations. It serves as a valuable resource for administrators who want to ensure the security of their vSAN deployments and safeguard their storage infrastructure and data.

NSX Security Configuration Guide

The VMware Cloud Foundation Operations Compliance Pack for NSX-T is updated to support the following standards:

- NSX-T 3.2 Security Configuration Guide
- NSX-T 3.1 Security Configuration Guide
- NSX-T 3.0 Security Configuration Guide
- **Note:** NSX-T is renamed to NSX.

For more details, download the file: [NSX Security Configuration Guide](#).

The NSX Security Configuration Guide is a comprehensive document provided by VMware that offers guidance and best practices for securing NSX, which is the next-generation network and security platform for software-defined networking.

The NSX Security Configuration Guide focuses on helping administrators secure the NSX infrastructure and protect network traffic, ensuring the confidentiality, integrity, and availability of data. It provides recommendations for various security considerations related to NSX, including authentication, authorization, network security, encryption, and compliance.

Here are some key areas covered in the NSX Security Configuration Guide:

- **NSX Component Hardening:** The guide provides recommendations for securing NSX components such as NSX nodes, NSX Controllers, and NSX Edge nodes. It includes guidelines for configuring strong authentication, implementing role-based access control (RBAC), and securing administrative access to NSX components.
- **Network Security:** It covers best practices for implementing network security within NSX. This includes recommendations for creating and managing security groups, implementing distributed firewalling, configuring security policies, and enforcing microsegmentation to control network traffic.
- **Authentication and Authorization:** The guide offers recommendations for configuring strong authentication mechanisms for NSX, such as integrating with identity providers, enabling multi-factor authentication, and defining access controls based on user roles and permissions.
- **Secure Network Communication:** It provides guidance for securing network communication within the NSX environment. This includes configuring Transport Layer Security (TLS) encryption for communication between NSX components, securing communication between NSX and vCenter, and protecting east-west and north-south traffic.

- **Virtual Machine Security:** The guide provides recommendations for securing virtual machines (VMs) deployed in NSX environments. This includes implementing security groups, defining security policies, and leveraging NSX's integration with third-party security solutions for advanced VM protection.
- **Logging and Auditing:** It emphasizes the importance of logging and auditing in NSX environments. The guide offers recommendations for enabling and configuring logging, integrating NSX with logging and monitoring solutions, and monitoring for security events and anomalies.
- **Compliance and Hardening:** The guide addresses compliance considerations and provides information on how NSX aligns with various security frameworks and regulations. It also offers recommendations for hardening NSX components and ensuring compliance with security policies.

The NSX Security Configuration Guide is regularly updated by VMware to incorporate the latest security best practices and recommendations. It serves as a valuable resource for administrators who want to ensure the security of their NSX Manager instances across their VMware Cloud Foundation platform, implement effective network security controls, and protect their software-defined networking infrastructure.

Regulatory Benchmark Details

Regulatory compliance benchmarks are standards or guidelines that help organizations measure and assess their level of compliance with applicable laws, regulations, and industry standards.

Note:

In future versions of VCF Operations, these benchmarks will be deprecated. A new, consolidated, out of the box benchmark called, "VCF 9 Security Baseline" will be introduced.

Health Insurance Portability and Accountability Act (HIPAA)

The Health Insurance Portability and Accountability Act (HIPAA) is a U.S. federal law enacted in 1996. HIPAA establishes standards and regulations to protect the privacy, security, and confidentiality of individuals' personal health information (PHI) and electronic health records (EHRs) in the healthcare industry.

The HIPAA Privacy Rule and Security Rule are two key components of the HIPAA standard:

- **HIPAA Privacy Rule:** The Privacy Rule sets standards for the use and disclosure of PHI by covered entities, which include healthcare providers, health plans, and healthcare clearinghouses. It grants individuals certain rights over their health information, such as the right to access, request amendments, and obtain an accounting of disclosures. Covered entities are required to implement safeguards to protect PHI, provide patients with notice of privacy practices, and obtain written authorization for certain uses and disclosures of PHI.
- **HIPAA Security Rule:** The Security Rule establishes security standards for protecting electronic PHI (ePHI). It requires covered entities and their business associates to implement administrative, physical, and technical safeguards to ensure the confidentiality, integrity, and availability of ePHI. These safeguards include risk assessments, access controls, encryption, audit controls, disaster recovery plans, and employee training on security awareness.

The HIPAA standard applies to healthcare providers, health plans, healthcare clearinghouses, and their business associates who handle PHI or ePHI. Compliance with HIPAA regulations is mandatory, and non-compliance can result in significant penalties, including financial fines and potential criminal charges.

HIPAA also includes provisions related to the electronic exchange of health information and establishes the Health Information Technology for Economic and Clinical Health (HITECH) Act, which promotes the adoption and meaningful use of electronic health records.

For the control set implemented by HIPAA, download the XLS file: [HIPAA Compliance](#).

Payment Card Industry Data Security Standard (PCI DSS) Compliance Standards

The Payment Card Industry Data Security Standard (PCI DSS) is a set of compliance standards established by the major payment card brands, including Visa, Mastercard, American Express, Discover, and JCB. PCI DSS aims to ensure the security of cardholder data and protect against fraud and unauthorized access within the payment card industry.

The PCI DSS compliance standards consist of twelve high-level requirements, organized into six control objectives. These requirements outline security measures that organizations handling payment card data must implement:

- Build and Maintain a Secure Network
- Protect Cardholder Data
- Maintain a Vulnerability Management Program.
- Maintain an Information Security Policy

PCI DSS compliance requirements vary based on the organization's level of involvement in payment card transactions, classified as levels 1 to 4. Level 1 merchants and service providers with the highest transaction volumes have more stringent requirements and undergo annual on-site audits by a Qualified Security Assessor (QSA). Level 2, 3, and 4 merchants may have different validation requirements, ranging from self-assessment questionnaires (SAQs) to external vulnerability scans.

Compliance with PCI DSS is necessary for entities involved in payment card processing, including merchants, service providers, and payment processors. Non-compliance may result in penalties, fines, increased transaction fees, or restrictions on card acceptance.

It's important to note that while this information provides an overview of the PCI DSS compliance standards, specific requirements and guidance can evolve over time. Therefore, it is recommended to consult the official PCI Security Standards Council (PCI SSC) website and the latest PCI DSS documentation for the most up-to-date information and requirements.

For the control set implemented, download the XLS file: [PCI DSS](#). Currently, Version 4.0 is supported.

CIS (Center for Internet Security) Security Standards

The VMware Aria Operations Compliance Pack for CIS is updated to support the following benchmarks:

- CIS_VMware_ESXi_6.7_Benchmark_V1.3.0
- CIS_VMware_ESXi_7.0_Benchmark_V1.2.0

The CIS (Center for Internet Security) Security Standards are a set of best practices and guidelines for securing computer systems and networks. The CIS organization is a non-profit entity that collaborates with experts from various industries to develop and promote consensus-based security configurations and benchmarks.

The CIS Security Standards include two primary components:

- **CIS Controls:** The CIS Controls are a set of 20 security actions that organizations can take to mitigate the most common and impactful cyber threats. These controls are prioritized based on their effectiveness in reducing risk. They cover various security domains, including asset management, access control, incident response, network security, and security awareness training. The CIS Controls are regularly updated to address emerging threats and evolving technology landscapes.
- **CIS Benchmarks:** CIS Benchmarks provide detailed configuration guidelines for securing specific technology platforms and systems. These benchmarks outline recommended settings and configurations for operating systems, applications, and network devices to ensure security and reduce vulnerabilities. CIS Benchmarks are created through a consensus-driven process involving input from cybersecurity experts, vendors, and practitioners.

CIS Security Standards are known for their practical and actionable nature, providing step-by-step instructions and specific configuration recommendations. They are widely adopted across industries and are used as a reference by organizations to assess, improve, and maintain the security of their IT systems and networks.

The CIS organization regularly updates its security standards and benchmarks to address emerging threats, technology advancements, and changes in regulatory requirements. The CIS Security Standards are available to the public, and organizations can leverage them as a valuable resource for enhancing their cybersecurity posture and reducing the risk of cyberattacks.

For the control set implemented, download the file: [CIS for VCF](#).

Defense Information Systems Agency (DISA) Security Standards

The Defense Information Systems Agency (DISA) establishes and provides security standards and guidelines for the U.S. Department of Defense (DoD) and its information systems. DISA is responsible for ensuring the secure operation and defense of DoD's global information infrastructure.

The VMware Aria Operations Compliance Pack for Defense Information Systems Agency (DISA) Security Standards is updated to the following version in VMware Aria Operations 8.18:

- VMware vSphere 7.0 STIG (U_VMW_vSphere_7-0_Y23M07_STIG V1R2 Dated 26 July 2023)
- VMware vSphere 8.0 STIG (U_VMW_vSphere_8-0_Y23M10_STIG Dated 31 October 2023)

DISA has developed several security standards and guidelines to protect sensitive information and ensure the integrity, availability, and confidentiality of DoD systems. Some of the key security standards and guidelines provided by DISA include:

- Security Technical Implementation Guides (STIGs): STIGs are a set of guidelines and configuration standards for various operating systems, applications, and network devices. They provide detailed instructions on how to secure and configure these systems to meet DoD security requirements. STIGs cover a wide range of technologies, including Windows, Linux, Cisco devices, databases, and web servers.
- Security Requirements Guides (SRGs): SRGs are comprehensive documents that outline security requirements for specific technology platforms, systems, or applications. They provide guidance on how to secure and configure systems in accordance with DoD security policies. SRGs address various security domains, including access control, identification and authentication, audit and accountability, and encryption.
- Security Technical Implementation Guides (STIGs) Viewer: DISA provides a STIG Viewer tool that helps organizations assess and implement STIG recommendations. The STIG Viewer automates the process of checking system configurations against STIG requirements, allowing organizations to identify and remediate security vulnerabilities more efficiently.
- Information Assurance Vulnerability Management (IAVM): DISA maintains the IAVM program, which identifies and manages vulnerabilities in DoD systems. IAVM alerts provide timely information about security vulnerabilities and patches. Organizations within the DoD are required to promptly apply these patches to mitigate potential risks.
- DoD Cybersecurity Discipline Implementation Plan (CDIP): The CDIP outlines the implementation and management of cybersecurity practices within the DoD. It provides guidelines and best practices for managing risks, protecting systems, responding to incidents, and fostering a culture of cybersecurity awareness.

DISA's security standards and guidelines play a critical role in ensuring the security and resilience of DoD systems and information assets. They are constantly updated and refined to address emerging threats and align with evolving cybersecurity practices. Organizations within the DoD are expected to adhere to these standards to maintain the security of their systems and networks.

For information about the control set implemented, download the file: [DISA](#).

The Federal Information Security Management Act (FISMA) Security Standards

The Federal Information Security Management Act (FISMA) is a U.S. federal law enacted in 2002. FISMA establishes a framework for securing information systems and managing cybersecurity risks within federal government agencies and their contractors. FISMA requires federal agencies to develop, implement, and maintain information security programs to protect sensitive government information.

While FISMA itself does not provide detailed security standards, it sets requirements for federal agencies to follow certain security guidelines and standards, including those established by the National Institute of Standards and Technology (NIST). NIST Special Publication (SP) 800-53, titled "Security and Privacy Controls for Federal Information Systems and Organizations," is a key document referenced under FISMA.

NIST SP 800-53 provides a catalog of security controls that federal agencies must implement to protect their information systems. The controls cover various areas, including access control, incident response, configuration management, encryption, network security, and security assessment and authorization. The controls are categorized into families and are tailored to address specific security requirements.

FISMA requires federal agencies to develop and maintain a risk-based approach to information security. This involves conducting risk assessments, implementing security controls based on the identified risks, periodically testing and evaluating the effectiveness of these controls, and ensuring continuous monitoring of information systems.

Under FISMA, federal agencies are also required to undergo annual security assessments, including independent audits, to evaluate the effectiveness of their information security programs and controls. The results of these assessments are reported to the Office of Management and Budget (OMB) and Congress.

FISMA compliance is crucial for federal agencies to demonstrate their commitment to protecting government information and ensuring the security of their information systems. It helps establish a standardized approach to information security across federal government entities and aligns with other security frameworks and standards, such as the NIST Cybersecurity Framework and NIST Risk Management Framework.

It's important to note that FISMA requirements may evolve over time, and agencies should refer to the latest guidance provided by NIST and other authoritative sources to ensure compliance with FISMA security standards.

International Organization for Standardization (ISO) Security Standards

The International Organization for Standardization (ISO) is an independent, non-governmental international standardization body that develops and publishes international standards across various industries. ISO has also created a series of security standards specifically related to information security management systems (ISMS). The most well-known among them is ISO/IEC 27001.

ISO/IEC 27001: The ISO/IEC 27001 standard specifies the requirements for establishing, implementing, maintaining, and continuously improving an ISMS within the context of an organization. It provides a systematic and risk-based approach to managing the security of sensitive information. The standard covers areas such as risk assessment, information security policies, asset management, access control, incident management, and compliance. ISO/IEC 27001 is widely adopted by organizations globally and serves as a benchmark for information security management.

ISO/IEC 27002: ISO/IEC 27002 (formerly known as ISO/IEC 17799) is a code of practice for information security controls. It offers guidance and recommendations for implementing security controls and safeguards based on the best practices of information security management. ISO/IEC 27002 covers a broad range of security areas, including organizational security, human resource security, physical and environmental security, communications and operations management, and compliance.

ISO/IEC 27005: ISO/IEC 27005 provides guidelines for conducting risk assessments in the context of information security. It offers a structured approach for identifying, analyzing, evaluating, and treating information security risks. ISO/IEC 27005 helps organizations assess the potential impact of risks, determine risk tolerance, and make informed decisions on implementing appropriate security controls.

ISO/IEC 27017 and ISO/IEC 27018: These standards specifically focus on cloud security. ISO/IEC 27017 provides guidelines for implementing information security controls in cloud computing environments, while ISO/IEC 27018 offers guidance for protecting personal data in the cloud and addresses privacy concerns related to cloud services.

ISO/IEC 27701: This standard is an extension to ISO/IEC 27001 and provides guidelines for implementing a Privacy Information Management System (PIMS). ISO/IEC 27701 helps organizations establish and maintain controls to protect personal data and comply with privacy regulations, such as the General Data Protection Regulation (GDPR).

ISO security standards provide a framework for organizations to establish effective information security management practices. Compliance with these standards demonstrates a commitment to securing sensitive information, managing risks, and implementing robust security controls. Organizations can seek ISO certification through a formal audit process conducted by accredited certification bodies to validate their adherence to ISO security standards.

For information about the control set implemented, download the file: [ISO](#). Currently, version ISO/IEC 27001:2022 is supported.

NIST SP 800-171

NIST Special Publication 800-171, titled "Protecting Controlled Unclassified Information in Nonfederal Systems and Organizations," provides a set of guidelines developed by the National Institute of Standards and Technology (NIST) to safeguard sensitive federal information residing in nonfederal systems. Specifically aimed at protecting Controlled Unclassified Information (CUI), the framework outlines 14 control families covering areas like access control, incident response, system integrity, and personnel security. It establishes standardized security requirements for contractors, vendors, and partners working with U.S. federal agencies, ensuring that CUI is consistently protected outside government networks. NIST SP 800-171 has become a cornerstone for compliance in industries such as defense, aerospace, and government contracting, often forming the basis for contractual cybersecurity obligations like those found in the DFARS (Defense Federal Acquisition Regulation Supplement).

For information about the control set implemented, download the file: [NIST SP 800-171](#).

NIST SP 800-53 R5

NIST Special Publication 800-53 Revision 5, titled "Security and Privacy Controls for Information Systems and Organizations," is a comprehensive framework developed by the National Institute of Standards and Technology (NIST) that provides an extensive catalog of security and privacy controls to protect federal information systems and organizations. Unlike NIST SP 800-171, which focuses on Controlled Unclassified Information in nonfederal systems, 800-53 R5 addresses a broader scope, covering federal systems, critical infrastructure, and even private-sector organizations that adopt it voluntarily. It organizes controls into 20 families, ranging from access control and risk assessment to incident response, supply chain protection, and privacy management. Revision 5 modernizes the framework by integrating privacy and security controls, emphasizing flexibility for diverse technologies, and reinforcing supply chain and system resilience. This publication plays a central role in federal cybersecurity strategy, forming the foundation for risk management frameworks like NIST RMF (Risk Management Framework).

For information about the control set implemented, download the file: [NISTSP 800-53](#).

Compliance Score Cards

The compliance page in VCF Operations displays score cards for each type of benchmark. A score card is a compliance visualization term.

What is a Compliance Score Card

Score cards in the Compliance landing page display the number of non-compliant objects, and the total number of objects affected by each hardening guide. A compliance score is displayed, which is counted as the ratio of compliant objects to total number of objects assessed by the given benchmark, represented in percentage. In addition, you can see the breakdown of the total number of objects that are compliant and non-compliant. You can click a score card to see more details, including alerts that were triggered based on the compliance standards.

The compliance score card of an object is counted as the smallest rounded off integer ($100 \times (\text{total number of symptoms triggered} / \text{total number of symptoms})$).

The compliance score for the object is based on the most critical of the violated standards. The score card displays 100 when all objects are compliant. When an object is non-compliant, the number of non-compliant symptoms are displayed in red and the total number of symptoms in grey.

Note: The compliance score for a user with limited object visibility is the same as for a user with complete object visibility. This is because the compliance score is calculated for all objects, irrespective of whether the user has access to the object or not.

Where You Find Compliance Score Cards

You can view score cards for each of the different types of benchmarks in the Compliance page available under **Security > Compliance**.

Compliance Page

In the **Security > Compliance** summary page, VCF Operations monitors compliance for VMware SDDC benchmarks.

VCF Operations displays compliance score cards for the following cards:

- VMware Cloud Foundation
- vCenter

Score cards are displayed for:

- VMware SDDC Benchmarks (for both data sources)
- VCF Benchmarks (for VMware Cloud Foundation)
- Custom Benchmarks (for both data sources)
- Regulatory Benchmarks (for both data sources)

Score Cards in the Compliance Page

In the **Security > Compliance** page, you can view scores for benchmarks that you have activated. Click a score card to view more information.

Table 1592: Compliance Page Score Card Options

Item	Description
Score card for the configured hardening guides, custom benchmark and management packs	Displays the compliance score, total compliant and non-compliant objects for the compliance standards you have configured.
Object Breakdown	Displays the number of compliant and non-compliant objects for the following types of objects: • vCenter • Host System • Virtual Machine • vSphere Distributed Port Group • vSphere Distributed Switch • vSAN Cache Disk • vSAN Capacity Disk • vSAN Cluster
Compliance Alert List	A list of alerts, grouped by time by default. You can either remove the grouping of the alerts, or group by criticality, definition, and object type. The alerts which caused the compliance violation are displayed in a table. You can sort the table by the following columns: • Alert ID • Criticality • Alert • Triggered On

Item	Description
	Updated On Click Actions to edit the policies applied to the configuration guide. Click an alert to view more details. The Alerts tab in the Inventory page opens. Use the advanced quick filter to search for alerts. Click the drop-down arrow for more options to narrow your search.

Compliance Alerts

You use the compliance score card as an investigative tool when you evaluate the state of objects in your VMware Cloud Foundation platform, or when you research the root cause of a problem. If the score card indicates a problem, you can view the alerts to see details about the violation. Violated rules are based on the symptoms defined in the compliance alert.

The compliance alerts, which have the subtype named Compliance, include one or more symptoms that represent the compliance rules. Compliance alerts that are triggered appear in the **Symptoms** tab in an object when you navigate to the **Alerts** tab in the Inventory section of VCF Operations. Compliance alerts appear as violations to the standard, and the triggered symptoms appear as violated rules. The rules are the alert symptoms, and the symptom configuration identifies the incorrect value or configuration. If a rule symptom is triggered for any of the alerts in the standard, the triggered rule violates the standard and affects the score.

How To Configure Compliance Benchmarks

Configure VMware SDDC, VCF, Custom and Regulatory benchmarks from the Compliance page available when you navigate to **Security > Compliance**. Unlike previous releases, you can now enable alert definitions in one of the active policies, from the Compliance page directly.

Activate VCF Benchmarks

You can activate the VCF benchmarks to audit the compliance of the objects in your VMware Cloud Foundation platform. The products currently included for assessment are ESX, SDDC Manager, vCenter, vSAN and NSX.

For VCF Operations to measure compliance against these data sources, you must first configure them. See the relevant topic in [Integrating Data Sources](#).

The Aria Operations CSA Compliance Pack is available as a Management Pack. Before you can enable the Aria Operations CSA Compliance Pack for VCF, you must install and activate the integration. Download the Management Pack from the [VMware Marketplace](#). For information on how to install a management pack, see the topic, [Adding and Managing Management Packs from VMware Marketplace](#).

- From the left menu, click **Security > Compliance** to access the compliance page.
- Select VMware Cloud Foundation from the carousel on the top of the page.
- Do the following:
 - To enable VCF Compliance based on the VCF Audit Guides, in the **VCF Benchmarks** section, click **Enable** in the **VCF Compliance based on the VCF Audit Guides** card.
 - To enable VCF Operations CSA Compliance Pack for VCF, in the **Regulatory Benchmarks** section, click **Enable** in the **VCF Compliance based on the Cloud Security Alliance** card. Ensure that you have installed the Aria Operations CSA Compliance Pack.
- Select the policy that you want to modify. When there are child policies, you can select a child policy and unselect a parent policy. VCF Operations modifies the selected policy and activates the alert definitions associated with the current scorecard.
- Click **Enable** to confirm your selection.

VCF Operations starts to assess the objects based on the policy that you selected. To edit a policy, click **Edit** in the configuration guide pane and select a different policy.

Activate VMware SDDC Benchmarks

You can activate the VMware SDDC Benchmark to monitor objects for violation of vSphere Security Configuration Guide, vSAN Security Configuration Guide, NSX Security Configuration Guide. The score cards in the VMware SDDC Benchmark warn you when compliance alerts trigger on your vCenter instance, NSX-V objects, NSX objects, vSAN objects, ESX hosts, virtual machines, distributed port groups, or distributed virtual switches.

You must configure the data source for which you are enabling the VMware SDDC benchmarks before you proceed. See the relevant topic in [Integrating Data Sources](#).

- From the left menu, click **Security > Compliance** to access the compliance page.
- Select a data source from the carousel on the top of the page.
- In the VMware SDDC Benchmarks section, click **Enable** under the vSphere Security Configuration Guide, vSAN Security Configuration Guide or NSX Security Configuration Guide card. The **Enable Policies** dialog box opens.
- Select the policy that you want to modify. When there are child policies, you can select a child policy and unselect a parent policy. VCF Operations modifies the selected policy and activates the alert definitions associated with the current scorecard.
- Click **Enable** to confirm your selection.

VCF Operations starts to assess the objects based on the policy that you selected. To edit a policy, click **Edit** in the configuration guide pane and select a different policy.

Create a New Custom Benchmark

You can create a custom compliance benchmark to ensure that objects comply with compliance alerts available in VCF Operations, or custom compliance alert definitions. When a compliance alert is triggered on your vCenter instance, ESX hosts, virtual machines, distributed port groups, or distributed switches, you investigate the compliance violation. You can add up to five custom compliance score cards.

To create a custom benchmark based on industry standard regulatory compliance requirements, you must first download and install the compliance management packs.

1. From the left menu, click **Security > Compliance** to access the compliance page.
2. In the Custom Benchmarks section, click **Add Custom Compliance**.
The **Add Custom Compliance** dialog box opens.
3. Select **Create a New Custom Benchmark**.
 - a) In the Name and Description step, provide a name and description for the custom benchmark and click **Next**.
 - b) In the Alert Definitions step, select the compliance alerts that you want to add to this custom compliance benchmark and click **Next**.
 - c) In the Policies step, select the policies to activate compliance and click **Finish**.

The custom compliance which monitors alert definitions that you selected is available in the Custom Benchmarks section of the Compliance page. You can edit the alert definitions and policies at any time by clicking **Edit**.

Import or Export a Custom Benchmark

You can export custom benchmarks from any VCF Operations instance and import it to another instance. Reusing custom benchmarks saves you time and effort. You can modify an imported custom benchmark. Exported files are in the XML format. The XML file contains information about alert groups, alerts, and filters.

You must first export a XML file with the custom benchmarks from another instance of VCF Operations before importing the XML file to another instance.

1. From the left menu, click **Security > Compliance** to access the compliance page.
2. Select a data source from the carousel on the top of the page.
3. In the Custom Benchmarks section, click **Add Custom Compliance**.
The **Add Custom Compliance** dialog box opens.
4. Select **Import An Existing Custom Benchmark**.
 - a) In the Import Compliance Score card dialog box, select the Score card Definition XML file from your local computer. If the XML file contains cloned alerts from the VCF Operations instance that was used to export the file, the cloned alerts are also imported.
 - b) VCF Operations displays a message to indicate if the XML file was successfully imported.
 - c) If you see a message which indicates that there is a conflict between the data in the XML file and the custom benchmarks already defined, make a selection on how to handle a conflict.
 - d) Click **Done**.
5. To export an existing custom benchmark, click the score card to select the benchmark and select **Export** from the **Actions** menu.

The imported compliance benchmarks are available in the Custom Benchmarks section of the Compliance page. You can edit the alert definitions and policies at any time by clicking **Edit** from the **Actions** menu after clicking the score card.

Activate a Regulatory Benchmark

To enforce and report on the compliance of your vSphere objects, you activate the compliance pack that contains the policies for regulatory standards. Then, you select the policy to activate the appropriate regulatory alerts for your virtual machines.

Before you can enable some of the regulatory benchmarks, you may need to install the integration by following the steps in [Adding and Managing Management Packs from VMware Marketplace](#) after downloading the .PAK file from [VMware Marketplace](#).

1. From the left menu, click **Security > Compliance** to access the compliance page.
2. To activate any regulatory benchmark, click **Activate From Repository** on the required compliance pack.
3. Click **Activate** to complete the installation.
4. To activate the compliance pack policies, navigate to the **Compliance** homepage and click **Enable** on the installed compliance pack.
The **Enable Policies** window opens.
5. Select the policies that you want to activate and click **Enable** to complete the process.

VCF Operations starts to assess the objects based on the regulatory benchmark that you installed.

Viewing and Configuring Audit Events

You can examine the diverse log events generated across resources in a VCF Instance, encompassing vSphere resources like vCenter, VMs, ESXhosts, vSAN resources, and NSX resources. By default, the Audit Events displays events from the past 24 hours, but you can change the display to view events in other time ranges. Each audit event has a detailed view containing information which helps you analyse the events in your virtual infrastructure for suspicious user actions or system issues.

What are Audit Events?

VCF Operations allows you to view, monitor and search through activities across all the vCenter instances in your VMware Cloud Foundation platform.

Audit events enhances operational efficiency, transparency, and accountability. These improvements facilitate security and compliance audits and provide essential details for conducting security forensics. Your organization may be required to provide audit data to support both internal and external audit reviews. By viewing audit events, you can reduce the time required for audit cycles by collecting data from various resources across vCenter instances, and see it in an unified interface. Currently, audit events reports on authentication and security-related audit events.

How Does Audit Logging Work?

VCF Operations can display audit events when you deploy the appliance VCF Operations for logs by using VCF Operations VCF Operations fleet management. If you have a VMware vSphere Foundation (VVF) license, you deploy VCF Operations for logs virtual appliance from an .ova file.

Audit events include interactions within the platform like searches, logins, logouts, capability checks, and configuration modifications, which result in the creation of relevant audit records. By default, the VCF Operations displays audit events for vCenter, VMs, Hosts, vSAN resources, NSX resources.

If your virtual infrastructure has multiple vCenter instances, VCF Operations, can monitor resources across all of them. Audit events are only displayed for vCenter instances which are monitored by VCF Operations for logs, even if such instances are directly configured with VCF Operations. You must configure monitoring of the vCenter instances in your virtual infrastructure. You can use log forwarding to configure multiple VCF Operations for logs instances to do the log analysis. Out of these instances, only one is integrated with VCF Operations, while the others forward the logs to the integrated VCF Operations for logs instance.

Note: The instances of VCF Operations for logs used for log forwarding must be of the supported version. Otherwise, audit events are not displayed for vCenter instances monitored by those vCenter log forwarders.

To summarize, VCF Operations displays audit events from vCenter instances which are:

- Configured in the VCF Operations for logs instance which is integrated with VCF Operations.
- Configured in the VCF Operations for logs instances which forward logs to the VCF Operations for logs which is integrated with VCF Operations.

Audit Event Categories

VCF Operations generates audit events for the following category of actions:

- Access
- Access_control
- Account_management
- Configuration
- Data_access
- Network
- Notification
- Permissions
- Session
- System
- Policy
- Firewall

Audit events may have more than one category.

The default time range for which the audit events are displayed is 24 hours. You have the option to select audit events from the past 1, 6, 12, 24 and 48 hours, but it depends on the data retention configured in the VCF Operations for logs instance. Additionally, you can specify a custom time range for any time period in the past, provided that it does not exceed the maximum limit of 48 hours.

Configuring Audit Events

Configure integration with VCF Operations for logs to see audit events from vCenter Servers.

To deploy the VCF Operations for logs appliance and configure monitoring of the vCenter instances, see one of the following topics:

- If you have a VCF license, see [Deploying VCF Operations for Logs](#).
- If you have a VMware vSphere Foundation (VVF) license, see [Deploying the VCF Operations for Logs Appliance for vSphere Foundation](#).
- To configure monitoring of the vCenter instances in your virtual infrastructure, see [Configuring a Account](#).

Viewing Audit Events

VCF Operations displays audit events to provide comprehensive visibility into vCenter instance activities, enabling monitoring and analysis of virtual infrastructure events for potential issues or security threats.

Where You Find Audit Events

From the left menu, click **Security > Audit Events** to access the Audit Events page. A list of audit events is displayed in a table in the right pane on the page. You can filter the audit events by using the options on the left pane. Click an event in the right pane to see more details. Use the search bar above the table to search for audit events. On the top of the page, you see the number of vCenter instances monitored.

Note: The audit events displayed on the page do not refresh automatically, when you are on the page. Use the **Reload** option to fetch the latest audit events.

Changing the Time Range

By default, VCF Operations displays audit events for the last 24 hours. You can change this time range by clicking the drop-down menu. The presets available are:

- Latest 1 hour
- Latest 6 hours
- Latest 12 hours
- Latest 24 hours
- Latest 48 hours

If you want to select a custom range, use the Start date, End date, Start Time and End time options. The custom range is limited to a 48 hour period. Click **Apply** to filter the audit events.

View Details

You can click the **View Details** link to view information about the vCenter instances the audit events are logged from.

The **vCenter Status** dialog box opens when you click the **View Details** link. Use the search bar to search for the monitoring status of specific vCenter instance.